Exercises

Level - 1

(Practice Questions Based on Fundamentals)

ABC of 3D-Geometry

1. Find the distance between the points P(3, 4, 5) and Q(–1, 2, –3).

2. Find the shortest distance of the point P(a, b, c) from x-axis.

3. If the line passing through the points (5, 1, a) and (3, b, 1) crosses the yz-plane at $\left(0, \frac{17}{2}, -\frac{13}{2}\right)$, find the value of a + b + 2.

4. Show that the points P(0, 7, 10), Q(–1, 6, 6) and C(–4, 9, 6) are the vertices of an isosceles rightangled triangle.

5. Find the locus of a point, the sum of distances from (1, 0, 0) and (–1, 0, 0) is 10.

6. Find the ratio in which the plane x – 2y + 3z = 17 divides the line joining the points (–2, 4, 7) and (3, –4, 8).

7. Find the co-ordinates of points which trisects the line joining the points p(–3, 2, 4) and Q(0, 4, 7).

8. If the centre of a tetrahedron OABC are (1, 2, –1), where A = (a, 2, 3), B = (1, b, 2), C = (2, 1, c) respectively, find the distance of P(a, b, c) from the orign.

9. Find the ratio in which the line joining the points P(–2, 3, 7) and Q(3, –5, 8) is divided by the yz-plane.

10. The vertices of a triangle are $A(5, 4, 6)$, $B(1, -1, 3)$ and $C(4, 3, 2)$. The internal bisector of $\angle BAC$ meets BC in D. Find AD.

Direction cosines and direction ratios

11. If a line is equally inclined ~~with~~ the axes, find its direction cosines.
12. Find the angle at which the vector $(4\mathbf{i} + 4\mathbf{j} + \mathbf{k})$ is inclined to the axes.
13. If a line makes an angle α, β, γ with the co-ordinate axes, find the value of

    (i) $\Sigma \sin^2 \alpha$ (ii) $\Sigma \cos(2\alpha)$.
14. A line OP makes with the x-axis and y-axis at an angle of 120°, 60°, respectively. Find the angle made by the line with the z-axis.
15. A vector $\mathbf{r}$ is inclined to x-axis and to y-axis at 60°. If $|\mathbf{r}| = 8$, find the vector $\mathbf{r}$.
16. Find the projection of the line joining (1, 2, 3) and (–1, 4, 2) on the line having direction ratios (2, 3, –6).
17. What is the angle between the lines whose direction cosines are $\left(-\frac{3}{4}, \frac{1}{4}, -\frac{\sqrt{3}}{2}\right)$ and $\left(-\frac{3}{4}, \frac{1}{4}, \frac{\sqrt{3}}{2}\right)$?
18. Find the angle between any two diagonals of a cube.
19. Find the angle between one diagonal of a cube and a diagonal of one face.
20. A line makes angles α, β, γ with the four diagonal of a cube, find the value of

    (i) $\Sigma \cos^2 \alpha$

    (ii) $\Sigma \sin^2 \alpha$

    (iii) $\Sigma \cos(2\alpha)$
21. Find the angle between the lines whose direction cosines are satisfy the equations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$.
22. Find the direction cosines of the two lines which are connected by the relation $l - 5m + 3n = 0$ and $7l^2 + 5m^2 - 3n^2 = 0$.
23. Find the angle between the lines which are connected by the relation $l + m + n = 0$ and $2lm + 2ln + mn = 0$.
24. Show that the straight lines whose direction cosines are given by the equations $al + bm + cn = 0$ and $ul^2 + vm^2 + wn^2 = 0$ are

    perpendicular if $a^2(u + v) + b^2(u + w) + c^2(u + v) = 0$

    and parallel if $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$.
25. If a variable line in two adjacent positions has direction cosines (l, m, n) and $(l + \delta l, m + \delta m, n + \delta n)$, show that the small angle $\delta\theta$ between the two positions is given by $(\delta\theta)^2 = (\delta l)^2 + (\delta m)^2 + (\delta n)^2$.

Striaght Line

26. Find the equation of a line passing through (2, 3, 4) and parallel to the line $\frac{x-3}{3} = \frac{y+1}{4} = \frac{z-7}{5}$.
27. Find the equation of a line passing through (–3, 2, –4) and parallel to the vector $2\mathbf{i} + 3\mathbf{j} + 7\mathbf{k}$.
28. Find the point where the line $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+3}{4}$ meets the plane $2x - 2y - z = 7$.
29. Find the equation of the line through $3x - 4y + 5z = 10$, $2x + 2y - 3z = 4$ and parallel to $x = 2y = 3z$..
30. Prove that the lines $4x = 3y = -z$ and $3x = y = -4z$ are perpendicular.
31. If the lines $x = az + b$, $y = cz + d$ and $x = a'z + b'$, $y = c'z + d'$ are perpendicular, prove that $a \cdot a' + c \cdot c' = -1$.
32. Find the distance of the point (1, –2, 3) from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$.
33. Find the foot of the perpendicular from the point (1, 2, 3) to the line $\frac{x-6}{3} = \frac{y-1}{2} = \frac{z-7}{3}$ and also find the length of the perpendicular.
34. Find the image of a point (1, 6. 3) in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-7}{3}$.
35. Find the equation of the line which can be drawn from the point (1, –1, 0) to intersect the lines $\frac{x-2}{2} = \frac{y-1}{3} = \frac{z-3}{4}$ and $\frac{x-4}{4} = \frac{y}{5} = \frac{z+1}{2}$. orthogonally.
36. Find the equation of the line through (2, –3, 1) parallel to the plane $2x - y + z = 6$ so as to meet the line $\frac{x-2}{2} = \frac{y}{-3} = \frac{z-2}{-1}$ a right angle.

Shortest distance between two lines

37. Find the shortest distance between the lines $\mathbf{r} = 3\mathbf{i} + 8\mathbf{j} + 3\mathbf{k} + \lambda(3\mathbf{i} - \mathbf{j} + \mathbf{k})$ and $\mathbf{r} = -3\mathbf{i} - 7\mathbf{j} + 6\mathbf{k} + \mu(-3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k})$.
38. Find the shortest distance between the lines

    $$L_1: \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$$

    and $$L_2: \frac{x-2}{2} = \frac{y+1}{-1} = \frac{z+2}{3}.$$
39. Prove that the shortest distance between any two opposite edges of a tetrahedron formed by the planes $y + z = 0$, $x + z = 0$, $x + y = 0$ and $x + y + z = \sqrt{3}a$ is $a\sqrt{2}$.

40. If $2d$ be the shortest distance between the lines $\frac{y}{b} + \frac{z}{c} = 1$, $x = 0$ and $\frac{x}{a} - \frac{z}{c} = \frac{1}{y} = 0$, prove that $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{d^2}$.

Plane

41. Find the equation of the plane through the points $P(1, 1, 1)$, $Q(3, -1, 2)$ and $R(-3, 5, -4)$.
42. Show that the four points (1, 2, 3), (2, 1, 4) (3, –1, 3) and (2, 2, 6) are coplanar.
43. Find the intercepts of the plane $2x - 4y + 5z = 20$ on the co-ordinate axes.
44. Find the equation of the plane passes through the points (3, 0, 0), (0, 4, 0) and (0, 0, 5).
45. Find the equations of the plane through the points (0, 4, –3), (6, –4, 3) other than the plane through the origin, which cut off from the axes intercepts, whose sum is zero,
46. A plane meets the co-ordinate axes in A, B, C such that the centroid of the triangle ABC is the point (p, q, r). Find the equation of the plane.
47. A variable plane cuts the co-ordinate axes in A, B, C and is of constant distance $3p$ from the origin. Find the locus of the centroid of the triangle ABC.
48. The plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ meets the co-ordinate axes at the points A, B, C respectively. Find the area of the ΔABC.
49. A variable plane cuts the co-ordinate axes in A, B, C and is of constant distance p from the origin. Find the locus of the centroid of the tetrahedron $OABC$.
50. If a variable plane forms a tetrahedron of constant volume $64k^3$ with the co-ordinate planes, find the locus of the centroid of the tetrahedron.
51. Find the angle between the planes $x - 2y + 2z + 2014 = 0$ and $2x + y + 2z + 2015 = 0$.
52. Prove that the planes $3x - 2y + 3z + 10 = 0$ and $2x - 3y - 4z + 7 = 0$ are perpendicular.
53. Find the equation of the plane passing through (1, 2, 3) and the perpendicular to the planes $2x + 3y + 4z + 7 = 0$, $3x + 4y + 5z + 10 = 0$
54. Find the angle between the line $\frac{x-2}{3} = \frac{y+1}{-1} = \frac{z-3}{-2}$ and the plane $3x + 4y + z + 5 = 0$.
55. Prove that the line $\frac{x-3}{2} = \frac{y-4}{3} = \frac{z-5}{4}$ is parallel to the plane $4x + 4y - 5z + 2 = 0$.
56. Find the equation of a plane passing through (2, 3, –1) and parallel to $x - 2y + 3z + 10 = 0$.
57. Find the equation of the plane passing through the points (2, 3, –4), (1, –1, 3) and parallel to x-axis.
58. Find the equation of the plane passing through the points (1, 2, 3), (4, 3, 1) and parallel to y-axis
59. Find the equation of the plane passing through the points (3, 4, 5), (–2, 3, –1) and parallel to z-axis.
60. Find the equation of the plane passing through (1, 2, 3) and parallel to xy-plane.
61. Find the equation of a plane parallel to yz-plane and passes through the point (2, 3, 4).
62. Find the equation of the plane passing through (3, 4, 5) and parallel to zx-plane.
63. Find the equation of the plane passing through the points (2, –1, 0), (3, –4, –5) and parallel to the line $2x = 3y = 4z$.
64. Find the equation of the plane passing through (2, 3, 4) and parallel to the lines $x = 2y = 3z$ and $2x = 5y = z$.
65. Find the equation of the plane passing through the intersection of the planes $2x + 5y - 5z = 6$ and $2x + 7y - 8z = 7$ and the point (–1, 4, 3).
66. Find the equation of the plane passing through the point (2, –1, 1) and the line $4x - 3y + 5 = 0 = y - 2z - 5$.
67. Find the equation of the plane passing through the intersection of the planes $2x + 3y + 10z = 8$ $2x - 3y + 7z = 2$ and the perpendicular to $3x - 2y + 4z = 5$.
68. Find the equation of the plane passing through the line $x + y + z + 3 = 0$, $2x - y + 3z + 1 = 0$ and parallel to $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$.
69. Find the length of the perpendicular from a point $P(1, 2, 3)$ to the plane $5x + 4y - 3z - 1 = 0$.
70. Find the equation of the line through (–1, 3, 2), perpendicular to the plane $x + 2y + 2z = 3$ and also find the co-ordinates of its foot.
71. Find the distance of the point (1, 2, 3) from the plane $x + y + z = 11$ measured parallel to $\frac{x+1}{1} = \frac{y+2}{-2} = \frac{z-7}{2}$.
72. Find the locus of a point, the sum of the squares of whose distances from the planes $x + y + z = 0$, $x - z = 0$, $x - 2y + z = 0$ is 9.
73. Two system of the co-ordinate axes have the same origin. If a plane cuts them at distances a, b, c and p, q, r respectively from the origin. Prove that $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{p^2} + \frac{1}{q^2} + \frac{1}{r^2}$.
74. Find the distance of the point (4, 1, 1) from the lines $x + y + z - 4 = 0 = x - 2y - 2z - 4$.
75. Find the distance between the planes $x + 2y - 2z + 1 = 0$ and $2x + 4y - 4z + 5 = 0$
76. Find the co-ordinates of the point of intersection of the line $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+3}{4}$ with the plane $2x - 2y - z = 7$.

77. Find the equation of the plane passing through (1, 2, 0) which contains the line $\frac{x+3}{3} = \frac{y-1}{4} = \frac{z-2}{-2}$.
78. Prove that the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ are coplanar and also find the equation of the plane where they lie.
79. Find the equation of the plane passing through the line of intersection of the planes $4x - 5y - 4z = 4$ and $2x + y + 2z = 8$ at the point (1, 2, 3).
80. Find the eqution of the plane bisecting the acute angle between the planes $x + 2y + 2z - 3 = 0$ and $3x + 4y + 12z + 1 = 0$.
81. Find the image of the point (2, −3, 4) with respect to the palne $4x + 2y - 4z + 3 = 0$.
82. Find the equation of the image of the line $\frac{x-1}{9} = \frac{y-2}{-1} = \frac{z+3}{-3}$ in the plane $3x - 3y + 10z - 26$.
83. Find the equation of the plane containning the line $\frac{x-1}{3} = \frac{y+6}{4} = \frac{z+1}{2}$ and is parallel to the line $\frac{x-4}{2} = \frac{y-1}{-3} = \frac{z+3}{5}$.
84. Find the equation of the plane containing the lines $\frac{x+1}{3} = \frac{y-1}{5} = \frac{z+3}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$.

Hints & Solution

4. Now, $PQ = \sqrt{1 + 1 + 16} = 3\sqrt{2}$

$QR = \sqrt{9 + 9 + 0} = 3\sqrt{2}$

and $RP = \sqrt{16 + 4 + 16} = \sqrt{36} = 6$

Thus, $PQ^2 + QR^2 = 18 + 18 = 36 = RP^2$

Therefore, the given triangle is right-angled isosceles.

5. Let the point be $P(x, y, z)$.

Consider $A = (1, 0, 0)$ and $B = (-1, 0, 0)$

Given $PA + PB = 10$

$\Rightarrow \sqrt{(x-1)^2 + y^2 + z^2} + \sqrt{(x+1)^2 + y^2 + z^2} = 10$

$\Rightarrow \sqrt{(x-1)^2 + y^2 + z^2} = 10 - \sqrt{(x+1)^2 + y^2 + z^2}$

$\Rightarrow (x-1)^2 + y^2 + z^2 = 100 + (x+1)^2 + y^2 + z^2 - 20\sqrt{(x-1)^2 + y^2 + z^2}$

$\Rightarrow (x-1)^2 - (x+1)^2 - 100 = -20\sqrt{(x+1)^2 + y^2 + z^2}$

$\Rightarrow -4x - 2 - 100 = -20\sqrt{(x-1)^2 + y^2 + z^2}$

$\Rightarrow -2x - 51 = -10\sqrt{(x-1) + y^2 + z^2}$

$\Rightarrow (2x + 51)^2 = 100((x-1)^2 + y^2 + z^2)$

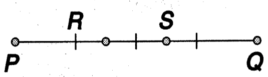
$\Rightarrow (4x^2 + 204x + 260) = 100(x^2 - 2x + 1 + y^2 + z^2)$

$\Rightarrow (96x^2 + 100y^2 + 100z^2) = (404x + 2501)$

which is the locus of the given point.

6. **

7. Let R divides P and Q internally in the ratio $1:2$.



Then the co-ordinates of R

$$= \left(\frac{0-6}{1+2}, \frac{4+4}{1+2}, \frac{7+8}{1+2}\right)$$

$$= \left(-2, \frac{8}{3}, 5\right)$$

Again, let S divides P and Q internally in the ratio $2:1$. Then the co-ordinates of S

$$= \left(\frac{0-3}{1+2}, \frac{8+2}{1+2}, \frac{14+4}{1+2}\right)$$

$$= \left(-1, \frac{10}{3}, 6\right).$$

8. We have,

$$\frac{a+1+2}{4} = 1 \Rightarrow a = 1$$

$$\frac{2+b+1}{4} = 2 \Rightarrow b = 5$$

and $$\frac{2+3+c}{4} = -1 \Rightarrow c = -9$$

Thus, the required distance from $P(a, b, c)$ to origin

$$= \sqrt{a^2 + b^2 + c^2}$$

$$= \sqrt{1 + 25 + 81} = \sqrt{107}.$$

9. Let the line PQ divides the xy-plane at R in the ratio $\lambda:1$.

Clearly, x-co-ordinate of the point R is zero.

Thus,

$$\frac{3\lambda - 2}{\lambda + 1} = 0$$

$\Rightarrow 3\lambda - 2 = 0$

$\Rightarrow \lambda = \frac{2}{3} = 2:3$

Therefore, the point R divides the yz-plane in the ratio $2:3$.

10. Now, $AB = \sqrt{4^2 + 5^2 + 3^2} = \sqrt{50} = 5\sqrt{2}$

and $AC = \sqrt{1 + 1 + 4^2} = \sqrt{81} = 3\sqrt{2}$

As we know that, if AD is the internal bisector of $\angle BAC$, then

$$\frac{BD}{DC} = \frac{AB}{AC}$$

$$\Rightarrow \frac{5\sqrt{2}}{3\sqrt{2}} = \frac{5}{3}$$

Thus, D divides BC internally in the ratio $5:3$, so therefore the co-ordinates of D

$$= \left(\frac{20+3}{8}, \frac{15-3}{8}, \frac{10+9}{8}\right)$$

$$= \left(\frac{23}{8}, \frac{12}{8}, \frac{19}{8}\right)$$

Thus, the length of AD

$$= \sqrt{\left(5 - \frac{23}{8}\right)^2 + \left(4 - \frac{7}{8}\right)^2 + \left(6 - \frac{19}{8}\right)^2}$$

$$= \sqrt{\left(\frac{17}{8}\right)^2 + \left(\frac{25}{8}\right)^2 + \left(\frac{26}{8}\right)^2}$$

$$= \sqrt{\frac{1755}{64}}$$

$$= \sqrt{\frac{1755}{8}}$$

11. Here, $\alpha = \beta = \gamma$

So, $\cos^2\alpha + \cos^2\alpha + \cos^2\alpha = 1$

$\Rightarrow 3\cos^2\alpha = 1$

$\Rightarrow \cos^2\alpha = \frac{1}{3}$

$\Rightarrow \cos\alpha = \pm\frac{1}{\sqrt{3}}$

Thus, the direction cosines of the line are

$$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \text{ or } \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right).$$

12. Let $\mathbf{r} = (4\mathbf{i} + 8\mathbf{j} + \mathbf{k})$ and then it makes angles α, β, γ with x, y and z axes respectively.

Let $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ unit vectors along the x, y and z axes respectively.

Now, $\quad \hat{\mathbf{r}} \cdot \hat{\mathbf{i}} = 4$

$\Rightarrow \quad |\mathbf{r}||\mathbf{i}| \cos\alpha = 4$

$\Rightarrow \quad \cos\alpha = \dfrac{4}{|\mathbf{r}|} = \dfrac{4}{9}$

$\Rightarrow \quad \alpha = \cos^{-1}\left(\dfrac{4}{9}\right)$

Smilarly, we can easily find that

$$\beta = \cos^{-1}\left(\frac{8}{9}\right), \ \gamma = \cos^{-1}\left(\frac{1}{9}\right)$$

13. we have

(i) $\Sigma\sin^2\alpha$

$= \sin^2\alpha + \sin^2\beta + \sin^2\gamma$

$= 1 - \cos^2\alpha + 1 - \cos^2\beta + 1 - \cos^2\gamma$

$= 3 - (\cos^2\alpha + \cos^2\beta + \cos^2\gamma)$

$= 3 - 1 = 2$

(ii) $\Sigma\cos(2\alpha)$

$= \cos(2\alpha) + \cos(2\beta) + \cos(2\gamma)$

$= 2\cos^2\alpha - 1 + 2\cos^2\beta - 1 + 2\cos^2\gamma - 1$

$= 2(\cos^2\alpha + \cos^2\beta + \cos^2\gamma) - 3$

$= 2 - 3 = -1$

14. Here, $\alpha = 120°$ and $\beta = 60°$

As we know that,

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + 1$

$\Rightarrow \quad \cos^2(120°) + \cos^2(60°) + \cos^2\gamma = 1$

$\Rightarrow \quad \left(-\dfrac{1}{2}\right)^2 + \left(\dfrac{1}{2}\right)^2 + \cos^2\gamma = 1$

$\Rightarrow \quad \cos^2\gamma = 1 - \left(\dfrac{1}{4} + \dfrac{1}{4}\right) = 1 - \dfrac{1}{2} = \dfrac{1}{2}$

$\Rightarrow \quad \cos\gamma = \pm\dfrac{1}{\sqrt{21}}$

$\Rightarrow \quad \gamma = \dfrac{\pi}{4}, \dfrac{3\pi}{4}$

15. Here, $l = \cos(60°) = m$

As we know that,

$l^2 + m^2 + n^2 = 1$

$\Rightarrow \quad \left(\dfrac{1}{2}\right)^2 + \left(\dfrac{1}{2}\right)^2 + n^2 = 1$

$\Rightarrow \quad n^2 = 1 - \left(\dfrac{1}{4}, \dfrac{1}{4}\right) = 1 - \dfrac{1}{2} = \dfrac{1}{2}$

$\Rightarrow \quad n = \pm\dfrac{1}{\sqrt{2}}$

Therefore,

$\mathbf{r} = |\mathbf{r}|(l\mathbf{i} + m\mathbf{j} + n\mathbf{k})$

$\Rightarrow \quad = 8\left(\dfrac{1}{2}\mathbf{i} + \dfrac{1}{2}\mathbf{j} \pm \dfrac{1}{\sqrt{2}}\mathbf{k}\right)$

$\Rightarrow \quad = (4\mathbf{i} + 4\mathbf{j} \pm 4\sqrt{2}\,\mathbf{k}).$

16. As we know that the projection of the segment joining the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on a line having direction cosines l, m, n is

$l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1).$

$= \dfrac{1}{7}(-1 - 1) + \dfrac{1}{7}(4 - 2) + \dfrac{1}{7}(2 - 3)$

$= -\dfrac{2}{7} + \dfrac{2}{7} - \dfrac{1}{7}$

$= -\dfrac{1}{7}$

17. Let θ be the angle between them. Then

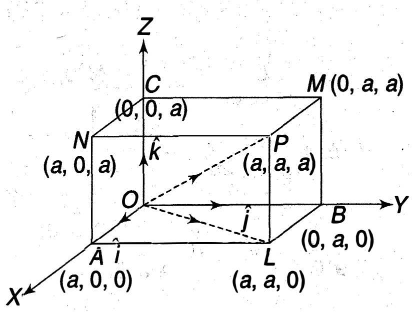
$\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$

$= \dfrac{9}{16} + \dfrac{1}{16} - \dfrac{3}{4}$

$= \dfrac{5}{8} - \dfrac{6}{8} = -\dfrac{1}{8}$

$\Rightarrow \quad \theta = \cos^{-1}\left(-\dfrac{1}{8}\right)$

18. Let $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are the unit vectors along the x, y and z axes, respectively.



Here, the four diangonals of a cube are

$\mathbf{OP} = a\mathbf{i} + a\mathbf{j} + a\mathbf{k}$

$\mathbf{AM} = -a\mathbf{i} + a\mathbf{j} + a\mathbf{k}$

$\mathbf{BN} = a\mathbf{i} - a\mathbf{j} + a\mathbf{k}$

$\mathbf{CL} = a\mathbf{i} + a\mathbf{j} - a\mathbf{k}$

Now, $\quad \mathbf{OP} \cdot \mathbf{AM} = -a^2 + a^2 + a^2$

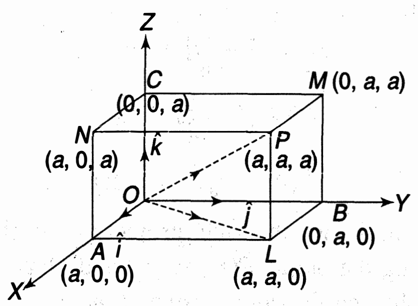
$\Rightarrow \quad |\mathbf{OP}||\mathbf{AM}|\cos\theta = a^2$

$\Rightarrow \quad a\sqrt{3}a\sqrt{3}\cos\theta = a^2$

$\Rightarrow \quad \cos\theta = \dfrac{a^2}{3a^2} = \dfrac{1}{3}$

Thus, $\quad \theta = \cos^{-1}\left(\dfrac{1}{3}\right)$

19. Let **i**, **j**, **k** are the unit vectors along the x, y and z axes, respectively.



Here, the diagonal of the cube = **OP** = a**i** + a**j** + a**k** and a diagonal of the face = **OL** = a**i** + a**j** + 0**k**

Thus, $\quad \mathbf{OP} \cdot \mathbf{OL} = a^2 + a^2 + 0$

$\Rightarrow \quad |\mathbf{OP}||\mathbf{OL}|\cos\varphi = 2a^2$

$\Rightarrow \quad a\sqrt{3}\, a\sqrt{2}\cos\varphi = 2a^2$

$\Rightarrow \quad \cos\varphi = \dfrac{2a^2}{\sqrt{2}\sqrt{3}a^2}$

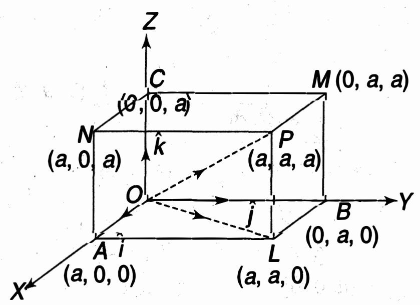
$= \dfrac{\sqrt{2}}{\sqrt{3}} = \sqrt{\dfrac{2}{3}}$

$\Rightarrow \quad \varphi = \cos^{-1}\left(\sqrt{\dfrac{2}{3}}\right)$

20. Let the direction cosines of a line be l, m, n, i.e.

r = l**i** + m**j** + n**k**

Let **i**, **j**, **k** are the unit vectors along the x, y and z axes, respectively.



Hence, the four diangonals of a cube are

OP = a**i** + a**j** + a**k**

AM = $-a$**i** + a**j** + a**k**

BN = a**i** − a**j** + a**k**

CL = a**i** + a**j** − a**k**

Then $\quad \cos\alpha = \dfrac{1}{\sqrt{3}}(l + m + n),$

$\cos\beta = \dfrac{1}{\sqrt{3}}(-l + m + n),$

$\cos\gamma = \dfrac{1}{\sqrt{3}}(l - m + n)$

and $\quad \cos\delta = \dfrac{1}{\sqrt{3}}(l + m - n)$

Now,

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta$

$= \dfrac{1}{3}(l + m + n)^2 + \dfrac{1}{3}(-l + m + n)^2$

$+ \dfrac{1}{3}(l - m + n)^2 + \dfrac{1}{3}(l + m - n)^2$

$= \dfrac{4}{3}(l^2 + m^2 + n^2)$

$= \dfrac{4}{3}$

21. We have

$l + m + n = 0$

$\Rightarrow \quad n = -1(l + m)$

Now, $\quad l^2 + m^2 - n^2 = 0$

$\Rightarrow \quad l^2 + m^2 = n^2 = (1 + m)^2$

$\Rightarrow \quad 2lm = 0$

$\Rightarrow \quad l = 0, m = 0$

When $\quad l = 0$, then $m = -n$

Thus, $\quad \dfrac{l}{0} = \dfrac{m}{-1} = \dfrac{n}{1}$

$\Rightarrow \quad \dfrac{l}{0} = \dfrac{m}{-1} = \dfrac{n}{1} = \dfrac{1}{\sqrt{2}}$

When $\quad m = 0$, then $l = -n$

Thus $\quad \dfrac{l}{-1} = \dfrac{m}{0} = \dfrac{n}{1}$

$\Rightarrow \quad \dfrac{l}{-1} = \dfrac{m}{0} = \dfrac{n}{1} = \dfrac{1}{\sqrt{2}}$

Let θ be the angle between them.

Then $\quad \cos\theta = l_1l_2 + m_1m_2 + n_1n_2$

$\Rightarrow \quad \cos\theta = 0 + 0 + \dfrac{1}{\sqrt{2}} \cdot \dfrac{1}{\sqrt{2}} = \dfrac{1}{2}$

$\Rightarrow \quad \theta = \dfrac{\pi}{3}$

22. We have,

$l - 5m + 3n = 0$

$\Rightarrow \quad l = 5m - 3n$

Now,

$7l^2 + 5m^2 + 3n^2 = 0$

$\Rightarrow \quad 7(5m - 3n)^2 + 5m^2 - 3n^2 = 0$

$\Rightarrow \quad 7(25m^2 + 9n^2 - 30mn) + 5m^2 - 3n^2 = 0$

$\Rightarrow \quad 180m^2 - 210mn + 60n^2 = 0$

$\Rightarrow \quad 6m^2 - 7mn + 2n^2 = 0$

$\Rightarrow \quad 6m^2 - 3mn - 4mn + 2n^2 = 0$

$\Rightarrow \quad 3m(2m - n) - 2n(2m - n) = 0$

$\Rightarrow \quad (3m - 2n)(2m - n) = 0$

$\Rightarrow \quad (3m - 2n) = 0, (2m - n) = 0$

Now,

$(3m - 2n) = 0$

$\Rightarrow \quad 3m = 2n$

$\Rightarrow \quad \frac{m}{2} = \frac{n}{3}$

$\Rightarrow \quad \frac{m}{2} = \frac{n}{2} = \frac{5m - 3n}{5.2 - 3.3} = \frac{l}{1}$

$\Rightarrow \quad \frac{l}{1} = \frac{m}{2} = \frac{n}{3} = \frac{1}{\sqrt{14}}$

Also, $\quad 2m - n = 0$

$\Rightarrow \quad \frac{m}{1} = \frac{n}{2}$

$\frac{5m - 3n}{5.1 - 3.2} = \frac{l}{-1}$

$\Rightarrow \quad \frac{l}{-1} = \frac{m}{1} = \frac{n}{2} = \frac{1}{\sqrt{6}}$

Hence, the direction cosines are

$\left(\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right)$ and $\left(-\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right)$.

23. We have,

$l + m + n = 0$

$\Rightarrow \quad n = -(l + m)$

Now,

$2lm + 2ln + mn = 0$

$\Rightarrow \quad 2lm + n(2l + m) = 0$

$\Rightarrow \quad 2lm - (l + m)(2l + m) = 0$

$\Rightarrow \quad (l + m)(2l + m) - 2lm = 0$

$\Rightarrow \quad (l^2 + m^2 + 3lm - 2lm) = 0$

$\Rightarrow \quad (l^2 + m^2 + lm) = 0$

$\Rightarrow \quad \left(\frac{l}{m}\right)^2 + \left(\frac{l}{m}\right) + 1 = 0$

Let its roots are $\frac{l_1}{m_1}, \frac{l_2}{m_2}$.

Thus, product of the roots = –1

$\Rightarrow \quad \frac{l_1}{m_1} \cdot \frac{l_2}{m_2} = -1$

$\Rightarrow \quad \frac{l_1 l_2}{1} = \frac{m_1 m_2}{-1}$

$\Rightarrow \quad \frac{l_1 l_2}{1} = \frac{m_1 m_2}{-1} = \frac{(l_1 l_2 + m_1 m_2)}{(1 - 1)}$

$\Rightarrow \quad \frac{l_1 l_2}{1} = \frac{m_1 m_2}{-1} = \frac{n_1 n_2}{0}$

$\Rightarrow \quad \frac{l_1 l_2}{1} = \frac{m_1 m_2}{-1} = \frac{n_1 n_2}{0} = \frac{1}{\sqrt{2}}$

Let θ be the angle between them.

Then $\quad \cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$

$= \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 0 = \theta$

$\Rightarrow \quad \theta = \frac{\pi}{2}$

24. We have,

$al + bm + cn = 0$

$\Rightarrow \quad l = -\frac{(bm + cn)}{a}$

Now, $\quad ul^2 + vm^2 + wn^2 = 0.$

$\Rightarrow \quad u\left(\frac{bm + cn}{a}\right)^2 + vm^2 + wn^2 = 0$

$\Rightarrow \quad u(bm + cn)^2 + va^2m^2 + wa^2n^2 = 0.$

$\Rightarrow \quad u(b^2m^2 + c^2n^2 + 2bcmn) + va^2m^2 + wa^2n^2 = 0$

$\Rightarrow \quad (ub^2 + va^2)m^2 + 2ubcmn + (uc^2 + wa^2)n^2 = 0$

$\Rightarrow \quad (ub^2 + va^2)\left(\frac{m}{n}\right)^2 + 2ubc\left(\frac{m}{n}\right) + (uc^2 + wa^2) = 0 \quad$ ...(i)

Let its roots are $\frac{m_1}{n_1}$ and $\frac{m_2}{n_2}$.

(a) If two straight lines are parallel, the Eq. (i) will provide us equal roots.

Thus, $D = 0$

$\Rightarrow \quad (2ubc)^2 - 4(b^2u + a^2v)(c^2u + a^2w) = 0$

$\Rightarrow \quad (ubc)^2 - (b^2u + a^2v)(c^2u + a^2w) = 0$

$\Rightarrow \quad (b^2u + a^2v)(c^2u + a^2w) - (ubc)^2 = 0$

$\Rightarrow \quad a^2b^2uw + a^2c^2uv + a^4vw = 0$

$\Rightarrow \quad b^2uw + c^2uv + a^2vw = 0$

$\Rightarrow \quad \frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$

(b) When two straight lines are perpendicular, the, product of the roots $= \frac{uc^2 + wa^2}{ub^2 + va^2}$

$\Rightarrow \quad \frac{m_1}{n_1} \cdot \frac{m_2}{n_2} = \frac{uc^2 + wa^2}{ub_2 + va^2}$

$\Rightarrow \quad \frac{m_1 m_2}{uc^2 + wa^2} = \frac{n_1 n_2}{ub^2 + va^2}$

Similarly, eliminating n, we get

$$\frac{l_1 l_2}{wb^2 + vc^2} = \frac{m_1 m_2}{uc^2 + wa^2}$$

Thus, $\quad \frac{l_1 l_2}{wb^2 + vc^2} = \frac{m_1 m_2}{uc^2 + wa^2} = \frac{n_1 n_2}{ub^2 + va^2}$

Two lines are perpendicular, if

$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$

$\Rightarrow \quad (wb^2 + vc^2) + (uc^2 + wa^2) + (ub^2 + va^2) = 0$

$\Rightarrow \quad a^2(u + v) + b^2(u + w) + c^2(u + v) = 0$

Hence, the result.

25. We have

$l^2 + m^2 + n^2 = 1$

and $(l + \delta l)^2 + (m + \delta m)^2 + (n + \delta n)^2 = 1$

$\Rightarrow \quad (\delta l)^2 + (\delta m)^2 + (\delta n)^2 + 2(l \cdot \delta l + m\delta m + n\delta n) = 0$

Now, $\delta\theta$ be the angle between two positions.

Then $\delta\theta = l(l + \delta l) + m(m + \delta m) + n(n + \delta n)$

$= 1 + (l \cdot \delta + m\delta m + n\delta n)$

$= 1 - \left(\frac{(\delta l)^2 + (\delta m)^2 + (\delta n)^2}{2}\right)$

$\Rightarrow \left(\frac{(\delta l)^2 + (\delta m)^2 + (\delta n)^2}{2}\right) = 1 - \cos(\delta\theta)$

$\Rightarrow (\delta l)^2 + (\delta m)^2 + (\delta n)^2 = 2(1 - \cos(\delta\theta))$

$= 4\left(\sin^2\left(\frac{\delta\theta}{2}\right)\right)$

$= 4 \cdot \left(\frac{\delta\theta}{2}\right)^2$

$\left(\text{since } \delta\theta \text{ is very small, } \Rightarrow \sin\left(\frac{\delta\theta}{2}\right) \to \left(\frac{\delta\theta}{2}\right)\right)$

$= (\delta\theta)^2$

$= (\delta n)^2$

26. Equation of the line passing through (2, 3, 4) and parallel to the given line is $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$

27. Equation of a line passing through (–3, 2, –4) and parallel to the given vector is $\frac{x+3}{2} = \frac{y-2}{3} = \frac{z+4}{7}$.

28. Any point on the given line can be considered as $(2\lambda + 1, 2 - 3\lambda, 4\lambda - 3)$. It is also a point on the plane. So,

$2(2\lambda + 1) - 2(2 - 3\lambda) - (4\lambda - 3) = 7$

$\Rightarrow 4\lambda + 6\lambda - 4\lambda = 7 - 2 + 4 - 3$

$\Rightarrow 6\lambda = 6$

$\Rightarrow \lambda = 1$

Hence, the co-ordinates of the plane is (3, –1, 1).

29. Equation of a line passing through (a, b, c) and parallel to $x = 2y = 3z$ is

$$\frac{x-a}{1} = \frac{y-b}{1/2} = \frac{z-c}{1/3}$$

which is passing through $3x - 4y + 5z = 10$, and $2x + 2y - 3z = 4$.

So, $3a - 4b + 5c = 0$

$2a + 2b - 2c = 0$

Solving, we get

$$\frac{a}{8-10} = \frac{b}{10+6} = \frac{c}{6+8}$$

$\Rightarrow \frac{a}{-2} = \frac{b}{16} = \frac{c}{14}$

$\Rightarrow \frac{a}{-1} = \frac{b}{8} = \frac{c}{7}$

Hence, the equation of the line is

$$\frac{x+1}{1} = \frac{y-8}{1/2} = \frac{z-7}{1/3}$$

30. The given lines $4x = 3y = -z$ and $3x = y = -4z$ can be written as

$L_1: \frac{x}{3} = \frac{y}{4} = \frac{z}{-12}$

$L_2: \frac{x}{4} = \frac{y}{-12} = \frac{z}{-3}$

Now, $a_1a_2 + b_1b_2 + c_1c_2 = 12 - 48 + 36 = 0$

Hence, the lines are perpendicular.

31. The given lines can be written as

$L_1: \frac{x-b}{a} = \frac{y-d}{c} = \frac{z}{1}$

and $L_2: \frac{x-b'}{a'} = \frac{y-d'}{c'} = \frac{z}{1}$

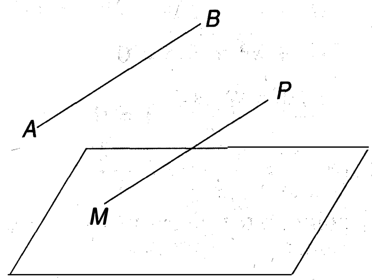
Since the given lines are perpendicular, so

$a \cdot a' + c \cdot c' + 1 = 0$

$\Rightarrow a \cdot a' + c \cdot c' = -1.$

32. Equation of any line passing through (1, –2, 3) and parallel to $\frac{x}{2} = \frac{y}{2} = \frac{z}{-6}$ is

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6}$$



Any point on the given line can be considered as $M(2\lambda + 1, 3\lambda - 2, 3 - 6\lambda)$ which is also a point (the plane. So,

$(2\lambda + 1) - (3\lambda - 2) + (3 - 6\lambda) = 5$

$\Rightarrow -7\lambda + 6 = 5$

$\Rightarrow \lambda = \frac{1}{7}$

Hence, the point M is $\left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$.

Hence, the required distance

$= \sqrt{\left(\frac{9}{5} - 1\right)^2 + \left(-\frac{11}{7} + 2\right)^2 + \left(\frac{15}{7} - 3\right)^2}$

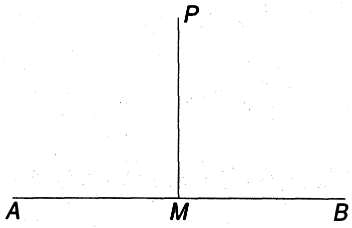
$= \sqrt{\frac{4}{49} + \frac{9}{49} + \frac{36}{49}}$

$= \sqrt{\frac{49}{49}}$

$= 1.$

33. Given line AB is

$$\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}.$$



Any point on the given line can be considered as $M(3\lambda + 6, 2\lambda + 7, 7 - 2\lambda)$.

Let the point P be (1, 2, 3).

Therefore, the direction ratios of PM

$$= (3\lambda + 6 - 1, 2\lambda + 7 - 2, 7 - 2\lambda - 3)$$

$$= (3\lambda + 5, 2\lambda + 5, 4 - 2\lambda)$$

Since PM is perpendicular to the given line, so

$$3(3\lambda + 5) + 2(2\lambda + 5) - 2(4 - 2\lambda) = 0$$

$\Rightarrow \quad 17\lambda + 17 = 0$

$\Rightarrow \quad \lambda = -1$

Thus, the point M is (3, 5, 9).

Therefore, the length of perpendicular

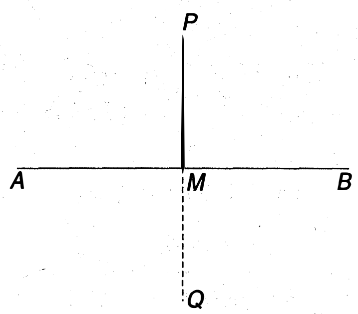
$$= \sqrt{(3-1)^2 + (5-2)^2 + (9-3)^2}$$

$$= \sqrt{4 + 9 + 36}$$

$$= \sqrt{49} = 7$$

34. Let $AB: \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

Any point on the given line AB is $(\lambda, 2\lambda + 1, 3\lambda + 2)$.



Now, the direction ratios of PM

$$= (\lambda - 1, 2\lambda + 1 - 6, 3\lambda + 2 - 3)$$

$$= (\lambda - 1, 2\lambda - 5, 3\lambda - 1)$$

Since PM is perpendicular to AB, so

$$(\lambda - 1) + 2(2\lambda - 5) + 3(3\lambda - 1) = 0$$

$\Rightarrow \quad 14\lambda = 14$

$\Rightarrow \quad \lambda = 1.$

Hence, the point M is (1, 3, 5).

Let $Q(\alpha, \beta, \gamma)$ be the image of P.

Therefore,

$$\frac{\alpha + 1}{2} = 1, \frac{\beta + 6}{2} = 3, \frac{\gamma + 3}{2} = 5$$

$\Rightarrow \quad \alpha = 1, \beta = 0, \gamma = 7$

Hence, the required image of (1, 6, 3) is (1, 0, 7).

35. The equation of any line passing through (1, –1, 0) with the direction ratios a, b and c is

$$\frac{x-1}{a} = \frac{y+1}{b} = \frac{z-0}{c} \quad \text{...(i)}$$

which is orthogonal to $\frac{x-2}{2} = \frac{y-1}{3} = \frac{z-3}{4}$

and $\quad \frac{x-4}{4} = \frac{y}{5} = \frac{z+1}{2}.$

So, $\quad 2a + 3b + 4c = 0$

and $\quad 4a + 5b + 2c = 0$

Solving, we get

$$\frac{a}{6-20} = \frac{b}{16-4} = \frac{c}{10-12}$$

$\Rightarrow \quad \frac{a}{-4} = \frac{b}{12} = \frac{c}{-2}$

$\Rightarrow \quad \frac{a}{2} = \frac{b}{-6} = \frac{c}{1}$

Hence, the equation of the required line is

$$\frac{x-1}{2} = \frac{y+3}{b} = \frac{z-1}{c}$$

36. The equation of any line passing through (2, –3, 1) with the direction ratios

$$\frac{x-2}{a} = \frac{y+3}{b} = \frac{z-1}{c} \quad \text{...(i)}$$

The line (i) is parallel to $2x - y + z = 6$

and perpendicular to $\frac{x-2}{2} = \frac{y}{-3} = \frac{z-2}{-1}$

So, $\quad 2a - b + c = 0$

and $\quad 2a - 3b - c = 0$

Solving, we get

$$\frac{a}{1+3} = \frac{b}{2+2} = \frac{c}{-6+2}$$

$\Rightarrow \quad \frac{a}{4} = \frac{b}{4} = \frac{c}{-4}$

$\Rightarrow \quad \frac{a}{1} = \frac{b}{1} = \frac{c}{-1}$

$\Rightarrow \quad \frac{x-1}{1} = \frac{y+3}{1} = 1 - z$

37. Here, $\quad \mathbf{r}_1 = 3\mathbf{i} + 8\mathbf{j} + 3\mathbf{k}$

$\mathbf{r}_2 = -3\mathbf{i} - 7\mathbf{j} + 6\mathbf{k}$

and $\quad \mathbf{u} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}$

$\mathbf{v} = -3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$

Now, $(\mathbf{r}_2 - \mathbf{r}_1) = -6\mathbf{i} - 15\mathbf{j} + 3\mathbf{k}$

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 1 \\ -3 & 2 & 4 \end{vmatrix} = -6\mathbf{i} - 15\mathbf{j} + 3\mathbf{k}$$

$$|\mathbf{u} \times \mathbf{v}| = \sqrt{36 + 225 + 9} = \sqrt{270} = 3\sqrt{30}$$

Also, $(\mathbf{r}_2 - \mathbf{r}_1)\cdot(\mathbf{u} \times \mathbf{v})$

$$= (-6\mathbf{i} - 15\mathbf{j} + 3\mathbf{k})\cdot(-6\mathbf{i} - 15\mathbf{j} + 3\mathbf{k})$$

$$= 36 + 225 + 9 = 270$$

Hence, the required shortest distance

$$= \frac{(\mathbf{r}_2 - \mathbf{r}_1)\cdot(\mathbf{u} \times \mathbf{v})}{|(\mathbf{u} \times \mathbf{v})|}$$

$$= \frac{270}{\sqrt{270}} = \sqrt{270}$$

38. Here, $\mathbf{r}_1 = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$

$$\mathbf{r}_2 = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$$

and $\mathbf{u} = 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$

$$\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$$

Now, $(\mathbf{r}_2 - \mathbf{r}_1) = -2\mathbf{j} - 2\mathbf{k}$

and $$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 4 & 5 \\ 2 & -1 & 3 \end{vmatrix}$$

$$= 17\mathbf{i} + \mathbf{j} - 11\mathbf{k}$$

$$\Rightarrow \quad |(\mathbf{u} \times \mathbf{v})| = \sqrt{289 + 1 + 121} = \sqrt{411}$$

Hence, the required shortest distance

$$= \frac{(\mathbf{r}_2 - \mathbf{r}_1)\cdot(\mathbf{u} \times \mathbf{v})}{|(\mathbf{u} \times \mathbf{v})|}$$

$$= \frac{20}{\sqrt{411}}.$$

39. The equation of the given lines

$$\mathbf{y} + \mathbf{z} = 0, \mathbf{x} + \mathbf{z} = 0, \mathbf{x} + \mathbf{y} = 0$$

$x + y + z = \sqrt{3}a$ can be written as

$$L_1: \frac{x}{1} = \frac{y}{1} = \frac{z}{-1}$$

$$L_2: \frac{x}{1} = \frac{y}{-1} = \frac{z}{-1}$$

Here, $\mathbf{u} = \mathbf{i} + \mathbf{j} - \mathbf{k}$

$$\mathbf{v} = \mathbf{i} - \mathbf{j}$$

Now, $$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & -1 \\ 1 & -1 & 0 \end{vmatrix}$$

$$= -\mathbf{i} - \mathbf{j} - 2\mathbf{k}$$

$$\Rightarrow \quad |(\mathbf{u} \times \mathbf{v})| = \sqrt{1 + 1 + 4} = \sqrt{6}$$

Also, $(\mathbf{r}_2 - \mathbf{r}_1) = \sqrt{3}a\mathbf{k}$

Hence, the required shortest distance

$$= \frac{(\mathbf{r}_2 - \mathbf{r}_1)\cdot(\mathbf{u} \times \mathbf{v})}{|(\mathbf{u} \times \mathbf{v})|}$$

$$= \frac{2\sqrt{3}a}{\sqrt{6}} = a\sqrt{2}$$

40. Given lines $\frac{y}{b} = \frac{z}{c} = 1, x = 0$ and $\frac{x}{a} - \frac{z}{c} = 1, y = 0$ can be written as

$$L_1: \frac{x}{0} = \frac{y - b}{b} = \frac{z}{-c}$$

and $$L_2: \frac{x - a}{a} = \frac{y}{0} = \frac{z}{c}$$

Here, $\mathbf{r}_1 = b\mathbf{j}$

$$\mathbf{r}_2 = a\mathbf{i}$$

and $\mathbf{u} = b\mathbf{j} - c\mathbf{k}$

$$\mathbf{v} = a\mathbf{i} + c\mathbf{k}$$

Now, $(\mathbf{r}_2 - \mathbf{r}_1) = a\mathbf{i} - b\mathbf{j}$

and $$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & b & -c \\ a & 0 & c \end{vmatrix}$$

$$= bc\mathbf{i} - ac\mathbf{j} - ab\mathbf{k}$$

$$\Rightarrow \quad |(\mathbf{u} \times \mathbf{v})| = \sqrt{(bc)^2 + (ac)^2 + (ab)^2}$$

Given shortest distance $= 2d$

$$\Rightarrow \quad \frac{(\mathbf{r}_2 - \mathbf{r}_1)\cdot(\mathbf{u} \times \mathbf{v})}{|(\mathbf{u} \times \mathbf{v})|} = 2d$$

$$\Rightarrow \quad \frac{abc + abc}{\sqrt{(bc)^2 + (ac)^2 + (ab)^2}} = 2d$$

$$\Rightarrow \quad \frac{2abc}{\sqrt{(bc)^2 + (ac)^2 + (ab)^2}} = 2d$$

$$\Rightarrow \quad \frac{abc}{\sqrt{(bc)^2 + (ac)^2 + (ab)^2}} = d$$

$$\Rightarrow \quad \frac{(abc)^2}{((bc)^2 + (ac)^2 + (ab)^2)} = d^2$$

$$\Rightarrow \quad \frac{1}{d^2} = \frac{((bc)^2 + (ac)^2 + (ab)^2)}{(abc)^2}$$

$$\Rightarrow \quad \frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

41. Hence, the equation of the plane is

$$\begin{vmatrix} x - 1 & y - 1 & z - 1 \\ 2 & -2 & 1 \\ -4 & 4 & -5 \end{vmatrix} = 0$$

$\Rightarrow \quad 6(x-1)+6(y-1)=0$

$\Rightarrow \quad x+y=2$

42. Hence, the equation of the plane is

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 1 & -1 & 1 \\ 2 & -3 & 0 \end{vmatrix} = 0$$

$\Rightarrow \quad 3(x-1)+2(y-2)-(z-3)=0$

$\Rightarrow \quad 3x+2y-z=4$

Now, $\quad 3.2+2.2-6-4=6+4-10=0$

Therefore, the given points are coplanar.

43. Given plane is $2x-4y+5z=20$

$$\Rightarrow \quad \frac{x}{10}+\frac{y}{-5}+\frac{z}{4}=1$$

which is the required intercept form of the plane.

44. The equation of any plane passing through (3, 0, 0), (0, 4, 0) and (0, 0, 5) is

$$\frac{x}{3}+\frac{y}{4}+\frac{z}{5}=1$$

45. Let the equation of the plane be

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1, \text{ where}$$

$$a+b+c=0 \qquad \text{...(i)}$$

which is passing through $(0, 4, -3)$, $(6, -4, 3)$. So

$$\frac{4}{b}-\frac{3}{c}=1$$

and $\quad \dfrac{6}{a}-\dfrac{4}{b}+\dfrac{3}{c}=1$

Solving, we get

$a=3, b=-2, c=-1$ or $a=3, b=6, c=-9$

Hence, the equation of the plane is

$$\frac{x}{3}-\frac{y}{2}-\frac{z}{1}=1$$

or $\quad \dfrac{x}{3}+\dfrac{y}{6}-\dfrac{z}{9}=1$

46. Let the equation of the plane be

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

Clealy, $\quad \dfrac{a}{3}=p, \dfrac{b}{3}=q, \dfrac{c}{3}=r$

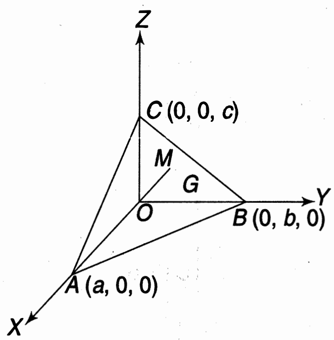
$\Rightarrow \quad a=3p, b=3q, c=3r$

Hence, the equation of the plane is

$$\frac{x}{p}+\frac{y}{q}+\frac{z}{r}=3$$

47. Let the equation of the plane be

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1.$$



Let the centroid G be (α, β, γ).

Clearly, $\quad a=3\alpha, b=3\beta, c=3\gamma$

It is given that, $OM=3p$

$$\Rightarrow \quad \left|\frac{0+0+0-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}\right|=3p$$

$$\Rightarrow \quad \left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right)=\frac{1}{9p^2}$$

$$\Rightarrow \quad \left(\frac{1}{9\alpha^2}+\frac{1}{9\beta^2}+\frac{1}{9\gamma^2}\right)=\frac{1}{9p^2}$$

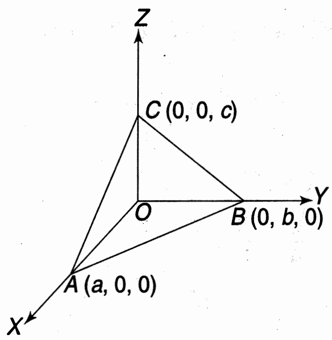
$$\Rightarrow \quad \left(\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2}\right)=\frac{1}{p^2}$$

Hence, the locus of the centroid is

$$\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)=\frac{1}{p^2}$$

48. Let the equation of the plane be

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$



Thus, the area of ΔABC

$$=\frac{1}{2}\left|(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a})\right|$$

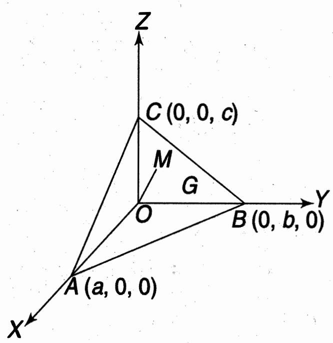
$$=\frac{1}{2}\left|(\vec{a}\hat{i}\times\vec{b}\hat{j}+\vec{b}\hat{j}\times\vec{c}\hat{k}+\vec{c}\hat{k}\times\vec{a}\hat{i})\right|$$

$$= \frac{1}{2}\,|(ab)\hat{k} + (bc)\hat{i} + (ca)\hat{j}|$$

$$= \frac{1}{2}\sqrt{(ab)^2 + (bc)^2 + (ca)^2}$$

49. Let the equation of the plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$



Let the centroid be $G(\alpha, \beta, \gamma)$

Clearly, $\frac{a}{4} = \alpha, \frac{b}{4} = \beta, \frac{c}{4} = \gamma$

$\Rightarrow \quad a = 4\alpha,\ b = 4\beta,\ c = 4\gamma$

It is given that $OM = p$

$$\Rightarrow \quad \left|\frac{0 + 0 + 0 - 1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}\right| = p$$

$$\Rightarrow \quad \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right) = \frac{1}{p^2}$$

$$\Rightarrow \quad \left(\frac{1}{(4\alpha)^2} + \frac{1}{(4\beta)^2} + \frac{1}{(4\gamma)^2}\right) = \frac{1}{p^2}$$

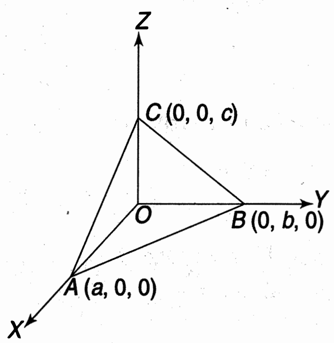
$$\Rightarrow \quad \left(\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}\right) = \frac{16}{p^2}$$

Hence, the locus of $G(\alpha, \beta, \gamma)$ is

$$\left(\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}\right) = \frac{16}{p^2}.$$

50. Let the equation of the plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$



It is given that, the volume of the tetrahedron

$$OABC = 64k^3$$

$$\Rightarrow \quad \frac{1}{6}\begin{vmatrix} 0 & 0 & 0 & 1 \\ a & 0 & 0 & 1 \\ 0 & b & 0 & 1 \\ 0 & 0 & c & 1 \end{vmatrix} = 64k^3$$

$$\Rightarrow \quad \frac{1}{6}(abc) = 64k^3$$

$$\Rightarrow \quad (abc) = 384\ k^3$$

Let the centroid be $G(\alpha, \beta, \gamma)$.

Clearly,

$$\frac{a}{4} = \alpha, \frac{b}{4} = \beta, \frac{c}{4} = \gamma$$

$$\Rightarrow \quad a = 4\alpha,\ b = 4\beta,\ c = 4\gamma$$

Therefore,

$$64(\alpha\,\beta\,\gamma) = 384k^3$$

$$\Rightarrow \quad (\alpha\,\beta\,\gamma) = 6k^3$$

Hence, the locus of $G(\alpha, \beta, \gamma)$ is

$$xyz = 6k^3$$

51. Let θ the angle between them. Thus,

$$\cos(\theta) = \frac{2 - 2 + 4}{\sqrt{1 + 4 + 4}\sqrt{4 + 1 + 4}}$$

$$= \frac{4}{9}$$

$$\Rightarrow \quad \theta = \cos^{-1}\left(\frac{4}{9}\right).$$

52. We have

$$a_1b_1 + a_2b_2 + a_3b_3$$

$$= 6 + 6 - 12$$

$$= 0$$

Thus, the given planes are perpendicular to each other.

53. Let the equation of the plane passing through (1, 2, 3,) is $a(x - 1) + b(y - 2) + c(z - 3) = 0$

which is perpendicular to the given planes

$2x + 3y + 4z + 7 = 0$ and $3x + 4y + 5z + 10 = 0$

So, $\quad 2a + 3b + 4c = 0$

and $\quad 3a + 4b + 3c = 0$

Solving, we get

$$\frac{a}{15 - 16} = \frac{b}{12 - 10} = \frac{c}{8 - 9}$$

$$\Rightarrow \quad \frac{a}{-1} = \frac{b}{2} = \frac{c}{-1}$$

$$\Rightarrow \quad \frac{a}{1} = \frac{b}{-2} = \frac{c}{1}$$

Hence, the equation of the plane is

$$(x - 1) - 2(y - 2) + (z - 3) = 0$$

$$\Rightarrow \quad x - 2y + z = 0$$

54. Do yourself.
55. Do yourself.
56. Equation of any plane parallel to $x - 2y + 3z + 10 = 0$ is $x - 2y + 3z + k = 0$ which is passing through (2, 3, –1). So,
$$2 - 6 - 3 + k = 0$$
$$\Rightarrow \quad k = 7$$
Hence, the equation of the plane is $x - 2y + 3z + 7 = 0$.
57. Let the equation of the plane be $by + cz + d = 0$. which is passing through the points (2, 3, –4) and (1, –1, 3).
Thus, $3b - 4c + d = 0$
and $b - c + d = 0$
Solving, we get
$$\frac{b}{-4+1} = \frac{c}{1-3} = \frac{d}{-3+4}$$
$$\Rightarrow \quad \frac{b}{-3} = \frac{c}{-2} = \frac{d}{1}$$
$$\Rightarrow \quad \frac{b}{3} = \frac{c}{2} = \frac{d}{-1}$$
Hence, the equation of the plane is
$$3y + 2z - 1 = 0$$
58. Do yourself
59. Do yourself.
60. The equation of any plane parallel to xy-plane is $z + k = 0$
which is passing through (1, 2, 3). So,
$$3 + k = 0$$
$$\Rightarrow \quad k = -3$$
Hence, the equation of the plane is $z = 3$.
61. Do yourself.
62. Do yourself.
63. Given line is
$$2x = 3y = 4z$$
$$\Rightarrow \quad \frac{2x}{12} = \frac{3y}{12} = \frac{4z}{12}$$
$$\Rightarrow \quad \frac{x}{6} = \frac{y}{4} = \frac{z}{3}$$
Hence, the required equation of the plane is
$$\begin{vmatrix} x-2 & y+1 & z \\ 1 & -3 & -5 \\ 6 & 4 & 3 \end{vmatrix} = 0$$
$$\Rightarrow \quad 11(x-2) - 33(y+1) + 22z = 0$$
$$\Rightarrow \quad (x-2) - 3(y+1) + 2z = 0$$
$$\Rightarrow \quad x - 3y + 2z = 5.$$
64. Given lines are
$$x = 2y = 3z \text{ and } 2x = 5y = z.$$
$$\Rightarrow \quad \frac{x}{6} = \frac{y}{3} = \frac{z}{2} \text{ and } \frac{x}{5} = \frac{y}{2} = \frac{z}{10}$$
Hence, the equation of the plane passing through (2, 3, 4) and parallel to the above lines is
$$\begin{vmatrix} x-2 & y-3 & z-4 \\ 6 & 3 & 2 \\ 5 & 2 & 10 \end{vmatrix} = 0$$
$$\Rightarrow \quad 22(x-2) - 50(y-3) - 3(z-4) = 0$$
$$\Rightarrow \quad 26x - 50y - 3z = 110.$$
65. The equation of any plane passing through the line of intersection of planes
$2x + 5y - 5z = 6$ and $2x + 7y - 8z = 7$ is
$(2x + 5y - 5z - 6) + \lambda(2x + 7y - 8z - 7) = 0$
which is passing through (–1, 4, 3).
$$\Rightarrow \quad (-2+20-15-6)+\lambda(-2+28-24-7)=0$$
$$\Rightarrow \quad -3 - 5\lambda = 0$$
$$\Rightarrow \quad \lambda = -\frac{3}{5}$$
Hence, the required equation of the plane is
$$(2x + 5y - 5z - 6) - \frac{3}{5}(2x + 7y - 8z - 7) = 0$$
$$\Rightarrow \quad 4x + 4y - z + 51 = 0.$$
66. The equation of any plane passing through the line of intersection of the planes $4x - 3y + 5 = 0$ and $y - 2z - 5 = 0$ is $(4x - 3y + 5) + \lambda(y - 2z - 5) = 0$ which is passing through (2, –1, 1) so,
$$\lambda = 1/2.$$
Hence, the equation of the required plane is
$$(4x - 3y + 5) + \frac{1}{2}(y - 2z - 5) = 0$$
$$\Rightarrow \quad 8x - 5y - 2z + 5 = 0$$
67. The equation of any plane passing through the line of intersection of the planes $2x + 3y + 10z = 8$ and $2x - 3y + 7z = 2$
is
$$(2x + 3y + 10z - 8) + \lambda(2x - 3y + 7z - 2) = 0$$
$$\Rightarrow \quad (2 + 2\lambda)x + (3 - 3\lambda)y + (10 + 7\lambda)z - (2\lambda + 8) = 0$$
which is perpendicular to $3x - 2y + 4z = 5$. So
$$3(2 + 2\lambda) - 2(3 - 3\lambda) + 4(10 + 7\lambda) = 0$$
$$\Rightarrow \quad 40 + 40\lambda = 0$$
$$\Rightarrow \quad \lambda = -1$$
Hence, the equation of the plane is
$$(2x + 3y + 10z - 8) - (2x - 3y + 7z - 2) = 0$$
$$\Rightarrow \quad 6y + 3z - 6 = 0$$
$$\Rightarrow \quad 2y + z - 2 = 0$$
68. Equation of any plane through the intersection of the planes $x + y + z + 3 = 0$, and $2x - y + 3z + 1 = 0$ is

$(x + y + z + 3) + \lambda(2x - y + 3z + 1) = 0$

$\Rightarrow \quad (1 + 2\lambda)x + (1 - \lambda)y + (3\lambda + 1)z + (\lambda + 3) = 0$

which is parallel to $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$.

So, $\quad (1 + 2\lambda) + 2(1 - \lambda) + 3(3\lambda + 1) = 0$

$\Rightarrow \quad 9\lambda - 6$

$\Rightarrow \quad \lambda = -\frac{2}{3}$

Hence, the equation of the plane is

$$(x + y + z + 3) - \frac{2}{3}(2x - y + 3z + 1) = 0$$

$\Rightarrow \quad 3(x + y + z + 3) - 2(2x - y + 3z + 1) = 0$

$\Rightarrow \quad -x + 5y - 3z + 7 = 0$

$\Rightarrow \quad x - 5y + 3z - 7 = 0.$

69. Hence, the length of the perpendicular from $P(1, 2, 3)$ to the plane $5x + 4y - 3z - 1 = 0$ is

$$\left|\frac{5 + 8 - 9 - 1}{\sqrt{25 + 16 + 9}}\right| = \frac{3}{5\sqrt{2}}$$

70. Equation of any line passing through (–1, 3, 2) and normal to the given plane is

$$\frac{x + 1}{1} = \frac{y - 3}{2} = \frac{z - 2}{2}$$

Any point on the above line can be written as $(\lambda - 1, 2\lambda + 3, 2\lambda + 2)$ which lies in the given plane.

So,

$$(\lambda - 1) + 2(2\lambda + 3) + 2(2\lambda + 2) = 3$$

$\Rightarrow \quad 9\lambda = -6$

$\Rightarrow \quad \lambda = -\frac{2}{3}$

Hence, the co-ordinates of the feet of the perpendicular are $\left(-\frac{5}{3}, \frac{5}{3}, \frac{2}{3}\right)$.

71. The equation of any line through (1, 2, 3) parallel to the given line is $\frac{x - 1}{1} = \frac{y - 2}{-2} = \frac{z - 3}{2}$.

Any point on the above line can be considered as $(\lambda + 1, 2 - 2\lambda, 2\lambda + 3$

which is also a common point of the plane.

Thus, $\quad \lambda + 1 + 2 - 2\lambda + 2\lambda + 3 = 11$

$\lambda = 5$

Therefore, the point is (6, –8, 13).

Hence, the required distance

$$= \sqrt{(6 - 1)^2 + (-8 - 2)^2 + (13 - 3)^2}$$

$$= \sqrt{25 + 100 + 100}$$

$$= \sqrt{225}$$

$$= 15$$

72. Let the point be $p(\alpha, \beta, \gamma)$

It is given that,

$$PM_1^2 + PM_2^2 + PM_3^2 = 9$$

$$\Rightarrow \left(\frac{\alpha + \beta + \gamma}{\sqrt{3}}\right)^2 + \left(\frac{\alpha - \gamma}{\sqrt{2}}\right)^2 + \left(\frac{\alpha - 2\beta + \gamma}{\sqrt{3}}\right)^2 = 9$$

$\Rightarrow \quad 2(\alpha + \beta + \gamma)^2 + 3(\alpha - \gamma)^2 + 2(\alpha - 2\beta + \gamma)^2 = 9$

$\Rightarrow \quad 7\alpha^2 + 10\beta^2 + 7\gamma^2 + 2\alpha\gamma - 4\alpha\beta - 4\beta\gamma = 9$

Hence, the locus of $p(\alpha, \beta, \gamma)$

$= 7x^2 + 10y^2 + 7z^2 + 2xz - 4xy - 4yz = 9.$

73. Let the equation of the planes be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ and $\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 1$

Let OM and ON be the distances from the origin to the given planes. Thus

$$OM = ON$$

$$\Rightarrow \left|\frac{0 + 0 + 0 - 1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}\right| = \left|\frac{0 + 0 + 0 - 1}{\sqrt{\frac{1}{p^2} + \frac{1}{q^2} + \frac{1}{r^2}}}\right|$$

$$\Rightarrow \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right) = \left(\frac{1}{p^2} + \frac{1}{q^2} + \frac{1}{r^2}\right)$$

Hence, the result.

74. Let a, b, c be the direction ratios of the line.

Since, the line is perpendicular to the normals to the plane, so

$$a + b + c = 0$$

and $\quad a - 2b - 2c = 0$

Solving, we get

$$\frac{a}{-2 + 2} = \frac{b}{1 + 2} = \frac{c}{-2 - 1}$$

$$\Rightarrow \quad \frac{a}{0} = \frac{b}{3} = \frac{c}{-3}$$

Let the point of intersection of the line with $z = 0$ plane.

Putting $z = 0$ in the given equations we get

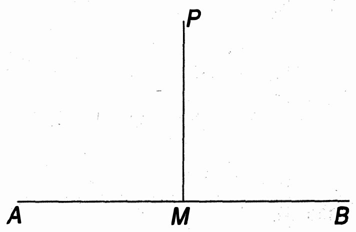
$$x + y = 4, x - 2y = 4$$

Solving, we get

$$x = 4, y = 0$$

Hence, the equation of the line is

$$\frac{x - 4}{0} = \frac{y}{3} = \frac{z}{-3}$$



Any point on the line be $M(4, 3\lambda, -3\lambda)$.

Now, the direction ratios of PM be $(0, 3\lambda - 1, -3\lambda - 1)$

Here, $PM \perp AB$

So, $0 + 3(3\lambda - 1) - 3(-3\lambda - 1) = 0$

$\Rightarrow \lambda = 0$

Hence, the point M is (4, 0, 0).

Therefore, the required distance

$$= \sqrt{0 + 1^2 + 1^2} = \sqrt{2}.$$

75. Hence the required distance between the planes

$$x + 2y - 2z + 1 = 0$$

and $2x + 4y - 4z + 5 = 0$

is $\left|\dfrac{(5/2) - 1}{\sqrt{1 + 4 + 4}}\right| = \dfrac{3/2}{3} = \dfrac{1}{2}$

76. Any point on the given line

$$\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+3}{4}$$

can be considered as $(2\lambda + 1, 2 - 3\lambda, 4\lambda - 3)$ which is the point of intersection. So it is also a point of the given plane.

Thus,

$$2(2\lambda + 1) - 2(2 - 3\lambda) - (4\lambda - 3) = 7$$

$\Rightarrow 6\lambda = 6$

$\Rightarrow \lambda = 1.$

Hence, the required point is (3, –1, 1).

77. Equation of any plane passing through (1, 2, 0) is

$$a(x - 1) + b(y - 2) + c(z - 0) = 0 \quad \text{...(i)}$$

which contains the given line. So,

$$a(-3 - 1) + b(1 - 2) + c(2 - 0) = 0$$

$\Rightarrow 4a + b - 2c = 0$ ...(ii)

and $3a + 4b - 2c = 0$ ...(iii)

Solving Eqs. (ii) and (iii), we get

$$\frac{a}{6} = \frac{b}{2} = \frac{c}{13}$$

Hence, the equation of the plane is

$$6(x - 1) + 2(y - 2) + 13(z - 0) = 0$$

$\Rightarrow 6x + 2y + 13z - 10 = 0.$

78. Now, $\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$

$$= \begin{vmatrix} 2-1 & 3-2 & 4-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 2 \\ 3 & 1 & 2 \end{vmatrix} \begin{pmatrix} C_2 \to C_2 - C_1 \\ C_3 \to C_3 - C_1 \end{pmatrix}$$

$= 0$

Hence, the given lines are coplanar.

Now, the equation of the plane where they lie is

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

$\Rightarrow -(x - 1) + 2(y - 2) - (z - 3) = 0$

$\Rightarrow -x + 2y - z = 0$

$\Rightarrow x - 2y + z = 0$

79. The Equation of any plane passing through the line of intersection of the planes

$4x - 5y - 4z = 4$ and $2x + y + 2z = 8$

is $(4x - 5y - 4z - 4) + \lambda(2x + y + 2z - 8) = 0$

which is passing through (1, 2, 3). Thus

$$2 - 22k = 0$$

$\Rightarrow k = \dfrac{1}{11}$

Hence, the equation of the plane is

$$(4x - 5y - 4z - 4) + \frac{1}{11}(2x + y + 2z - 8) = 0.$$

$\Rightarrow 13x + 3 \cdot y + 9z = 46.$

80. The given planes can be written as

$$-x - 2y - 2z + 3 = 0$$

and $3x + 4y + 12z + 1 = 0$

Now, $a_1a_2 + b_1b_2 + c_1c_2 = -3 - 8 - 24 < 0$

Thus, the acute angle bisector is

$$\left(\frac{-x - 2y - 2z + 3}{\sqrt{9}}\right) = -\left(\frac{3x + 4y + 12z + 1}{\sqrt{169}}\right)$$

$\Rightarrow 11x + 19y + 31z = 18.$

81. The equation of a line passing through (2, –3, 4) and normal to a plane is

$$\frac{x-2}{4} = \frac{y+3}{2} = \frac{z-4}{-4} \quad \text{....(i)}$$

Any point on Eq. (i) is

$$M(4\lambda + 2, 2\lambda - 3, 4 - 4\lambda)$$

Since the point M lies on the plane, so

$$4(4\lambda + 2) + 2(2\lambda - 3) - 4(4 - 4\lambda) + 3 = 0$$

$\Rightarrow 36\lambda = 11$

$\Rightarrow \lambda = \dfrac{11}{36}$

Hence, the M point $= \left(\dfrac{29}{9}, -\dfrac{33}{18}, \dfrac{25}{9}\right)$

Let $Q(\alpha, \beta, \gamma)$ be the image of $P(2, -3, 4)$.

Here, M is the mid-point of P and Q.

Thus, $\dfrac{\alpha + 2}{2} = \dfrac{29}{9}, \dfrac{\beta - 3}{2} = \dfrac{-33}{18}, \dfrac{\gamma + 4}{2} = \dfrac{25}{9}$

$\Rightarrow (\alpha, \beta, \gamma) = \left(\dfrac{40}{9}, -\dfrac{2}{3}, \dfrac{14}{9}\right)$

which is the required image of $P(2, -3, 4)$.

82. Given line is

$$\frac{x-1}{9}=\frac{y-2}{-1}=\frac{z+3}{-3} \quad ....(i)$$

and the plane is

$$3x-3y+10z=26. \quad ...(ii)$$

The direction ratios of the line are 9, –1, –3 and the direction ratios of the normal to the given plane are 3, –3 and 10.

Clearly line (i) is parallel to the plane (ii).

Let Q be the inage of the point $P(1, 2, -3)$.

Consider $Q=(\alpha, \beta, \gamma)$

Now,

$$\frac{\alpha-1}{3}=\frac{\beta-2}{-3}=\frac{\gamma+3}{10}=-2\left(\frac{3-6-30-26}{9+9+100}\right)$$

$$\frac{\alpha-1}{3}=\frac{\beta-2}{-3}=\frac{\gamma+3}{10}=1$$

$\alpha=4, \beta=-1, \gamma=7$

Hence, the required image of the line w.r.t. the given plane is

$$\frac{x-4}{9}=\frac{y+1}{-1}=\frac{z-7}{-3}$$

83. Any plane containing the line

$$\frac{x-1}{3}=\frac{y+6}{4}=\frac{z+1}{2} \text{ is}$$

$$a(x-1)+b(y+6)+c(z+1)=0 \quad ...(i)$$

where $\quad 3a+4b+2c=0 \quad ...(ii)$

Also, it is parallel to the second line. So,

$$2a-3b+2c=0 \quad ...(iii)$$

Solving Eq. (ii) and (iii), we get

$$\frac{a}{26}=\frac{b}{-11}=\frac{c}{-17}$$

Hence, the equation of the plane is

$$26(x-1)-11(y+6)-17(z+1)=0$$

$\Rightarrow \quad 26x-11y-17z=109$

84. The equation of the line containg the givel two lines is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}=0$$

$$\Rightarrow \quad \begin{vmatrix} x+1 & y+3 & z+5 \\ 3 & 5 & 7 \\ 1 & 3 & 5 \end{vmatrix}=0$$

$\Rightarrow \quad 4(x+1)-8(y+3)+4(z+5)=0$

$\Rightarrow \quad 4x-8y+4z=0$

$\Rightarrow \quad x-2y+z=0$